

Global Innovation Index 2025



Innovation performance of ASEAN in the GII 2025

The GII 2025 covers 10 economies in ASEAN: Brunei Darussalam, Indonesia, Cambodia, Lao People's Democratic Republic, Myanmar, Malaysia, Philippines, Singapore, Thailand and Viet Nam.



Total population
693.7M



Total GDP (PPP, intl. dollars)
1,842,853.2bn

GII 2025– Science and Technology Investment in ASEAN compared to the World

	Scientific publications	R&D investments *	Venture capital	PCT Patents
ASEAN (short term)	▲ 5.2% 2023-2024	▲ 3% 2022-2023	▼ -13.9% 2023-2024	▼ -0.9% 2023-2024
World benchmark (short term)	▲ 5.6% 2023-2024	▲ 4.4% 2022-2023	▼ -4.4% 2023-2024	▲ 0.5% 2023-2024

* R&D expenditures for 2024 are WIPO estimates for all economies, based on the latest available R&D intensity data and estimations of GDP growth at constant prices from the International Monetary Fund, World Economic Outlook Database, April 2025.

	Scientific publications	R&D investments	Venture capital	PCT Patents
ASEAN (long term)	▲ 6.7% 2015-2024	▲ 7.6% 2013-2023	▼ -3% 2020-2024	▲ 5.6% 2015-2024
World benchmark (long term)	▲ 4.2% 2015-2024	▲ 5% 2013-2023	▲ 0.4% 2020-2024	▲ 2.6% 2015-2024

Note: Long-term annual growth refers to the compound annual growth rate (CAGR) over the indicated period. Historic data may have been updated and might differ from last year's Global Innovation Tracker. Figures are rounded. Estimates or incomplete data are indicated by an asterisk (*). n.a. indicates not available.

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GII 2025 Rankings of economies in ASEAN

Economy	Economy Regional rank	GII rank	Innovation Input rank	Innovation Output rank
Singapore	1	5	1	9
Malaysia	2	34	30	39
Viet Nam	3	44	50	37
Thailand	4	45	46	43
Philippines	5	50	59	49
Indonesia	6	55	60	59
Brunei Darussalam	7	88	47	134
Cambodia	8	100	88	105
Lao People's Democratic Republic	9	109	83	129
Myanmar	10	122	131	99

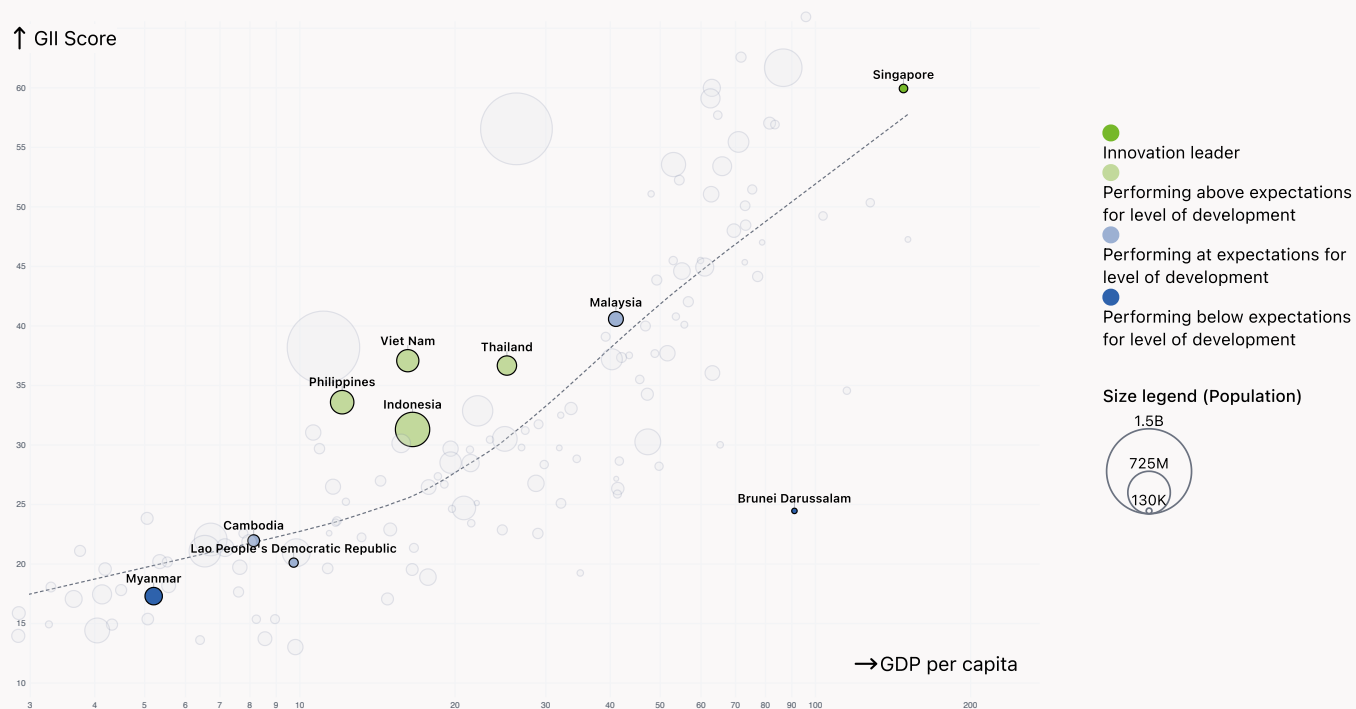
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Expected vs. Observed Innovation Performance in ASEAN

The bubble chart below shows the relationship between income levels (GDP per capita) and innovation performance (GII score). The trend line gives an indication of the expected innovation performance according to income level. Economies appearing above the trend line are performing better than expected and those below are performing below expectations.

> Innovation overperformers relative to their economic development



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Effectively translating innovation investments into innovation outputs in ASEAN

The chart below shows the relationship between innovation inputs and innovation outputs. Economies above the line are effectively translating costly innovation investments into more and higher-quality outputs.

> Relationship between innovation inputs and outputs



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Benchmark of ASEAN against other economy groupings for each of the seven areas of the GII Index

Institutions

Top 10 | Score: 78.63

Top 50 | Score: 66.62

Top 70 | Score: 61.59

ASEAN | Score: 55.28

Human capital and research

Top 10 | Score: 59.30

Top 50 | Score: 47.61

Top 70 | Score: 42.78

ASEAN | Score: 30.31

Infrastructure

Top 10 | Score: 61.36

Top 50 | Score: 55.76

Top 70 | Score: 52.46

ASEAN | Score: 42.52

Market sophistication

Top 10 | Score: 61.82

Top 50 | Score: 50.04

Top 70 | Score: 46.32

ASEAN | Score: 44.13

Business sophistication

Top 10 | Score: 59.10

Top 50 | Score: 45.25

Top 70 | Score: 40.43

ASEAN | Score: 32.17

Knowledge and technology outputs

Top 10 | Score: 54.93

Top 50 | Score: 37.97

Top 70 | Score: 33.31

ASEAN | Score: 23.17

Creative outputs

Top 10 | Score: 55.98

Top 50 | Score: 42.80

Top 70 | Score: 37.58

ASEAN | Score: 24.28

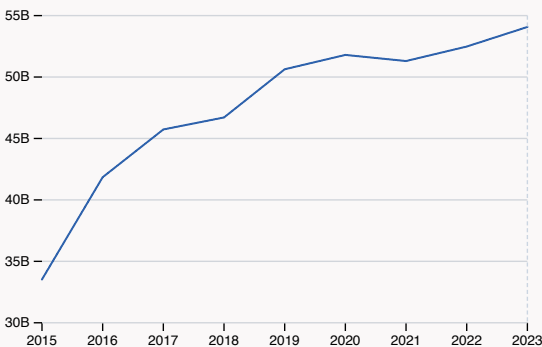
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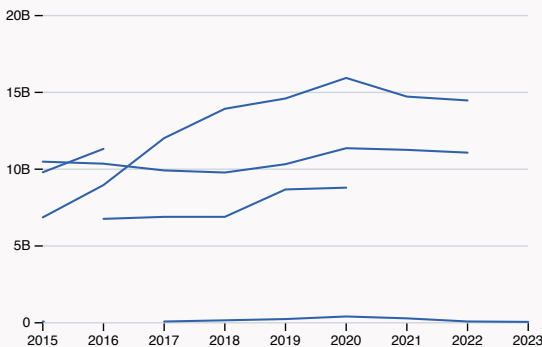
Selected Innovation Trends in ASEAN

> Innovation inputs

Regional value



Economy values



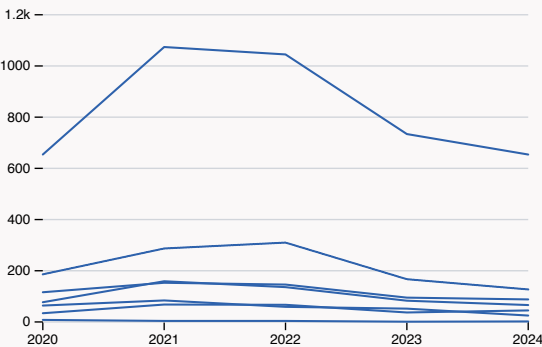
2.3.2 Gross Expenditure on R&D (GERD)

ASEAN's Gross Expenditure on R&D (GERD) changed at an annual rate of 6% (CAGR) between 2015 and 2023, rising from 33.53B USD to 54.07B USD. In 2023, GERD was up by 3.0 percentage points compared to the year prior.

Regional value



Economy values



4.2.2 Venture Capital (VC) Received

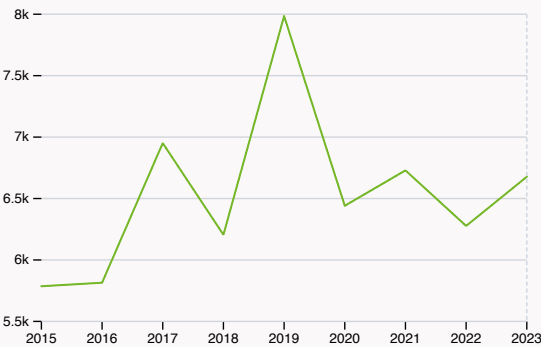
ASEAN's Venture Capital (VC) Received grew at an annual rate of -3% (CAGR) between 2020 and 2024, falling from 1.14k deals to 1.01k deals. In 2024, VC deals were down by -13.9 percentage points compared to the year prior.

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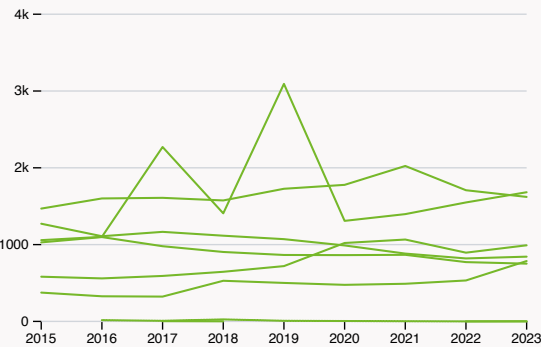


> Innovation outputs

Regional value



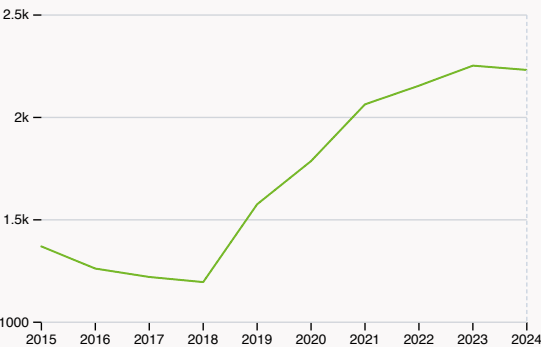
Economy values



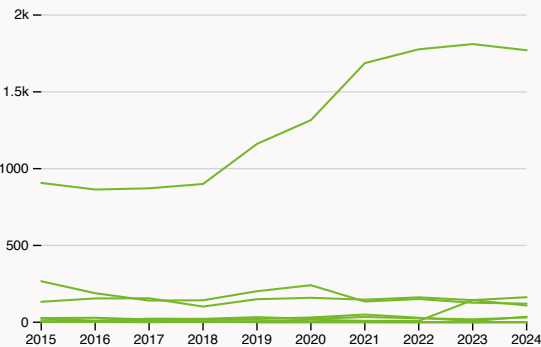
6.1.1 Patents by origin

The number of Patents by origin in ASEAN grew at an annual rate of 2% (CAGR) between 2015 and 2023, rising from 6k patents to 7k patents. In 2023, Patents by origin were up by 6.4 percentage points compared to the year prior.

Regional value



Economy values



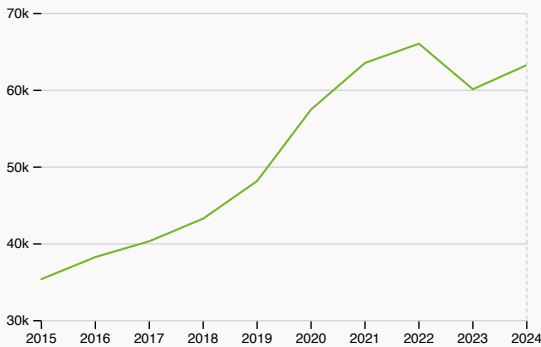
6.1.2 PCT patents by inventor origin

The number of PCT patents by inventor origin in ASEAN grew at an annual rate of 6% (CAGR) between 2015 and 2024, rising from 1k PCT patents to 2k PCT patents. In 2024, PCT patents by inventor origin were down by -0.9 percentage points compared to the year prior.

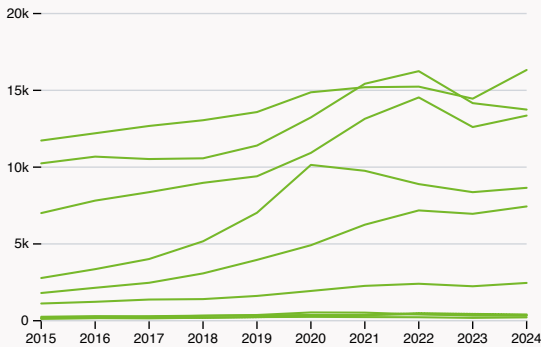
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Regional value



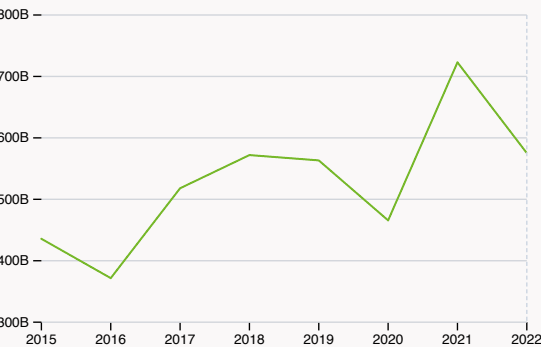
Economy values



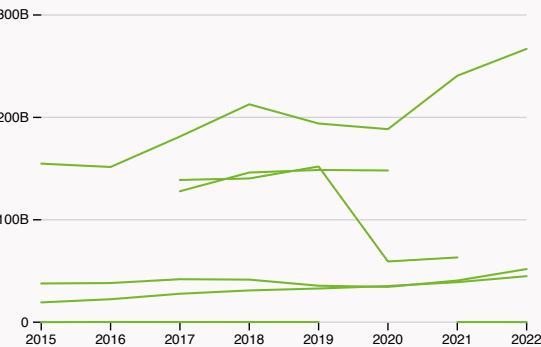
6.1.4 Scientific and technical articles

The number of Scientific and technical articles in ASEAN grew at an annual rate of 7% (CAGR) between 2015 and 2024, rising from 35k articles to 63k articles. In 2024, Scientific and technical articles were up by 5.2 percentage points compared to the year prior.

Regional value



Economy values



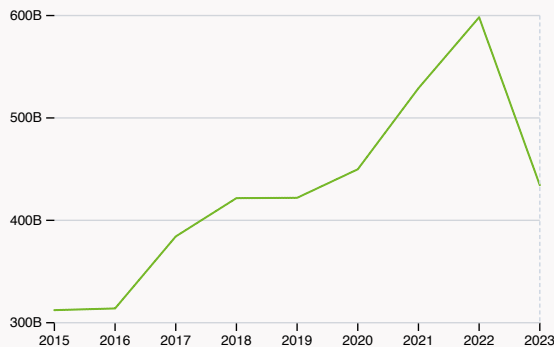
6.2.4 High-tech manufacturing

The value of High-tech manufacturing in ASEAN grew at an annual rate of 4% (CAGR) between 2015 and 2022, rising from USD 436 billion to USD 575 billion. In 2022, High-tech manufacturing was down by -20.4 percentage points compared to the year prior.

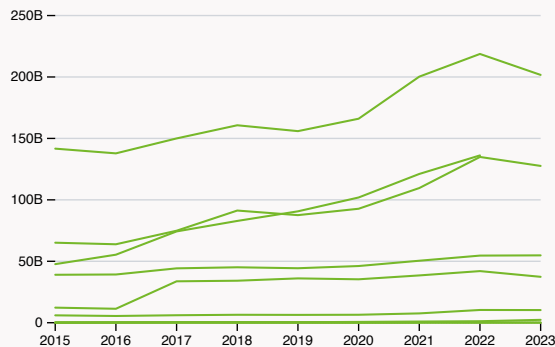
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Regional value



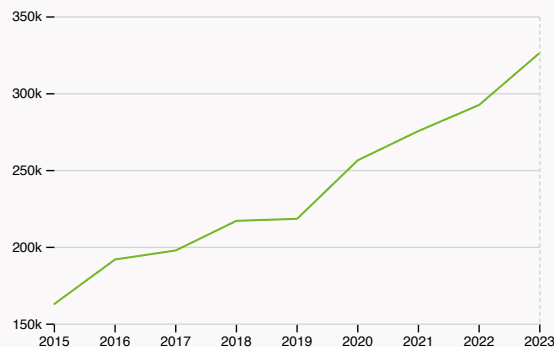
Economy values



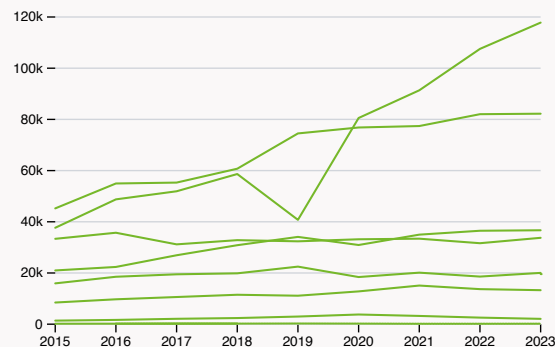
6.3.3 High-tech exports

The value of High-tech exports in ASEAN grew at an annual rate of 4% (CAGR) between 2015 and 2023, rising from USD 312 billion to USD 434 billion. In 2023, High-tech exports were down by -27.4 percentage points compared to the year prior.

Regional value



Economy values



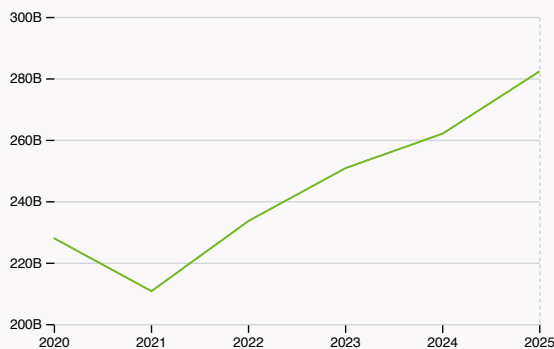
7.1.2 Trademarks by origin

The number of Trademarks by origin in ASEAN grew at an annual rate of 9% (CAGR) between 2015 and 2023, rising from 163k trademarks to 327k trademarks. In 2023, Trademarks by origin was up by 11.6 percentage points compared to the year prior.

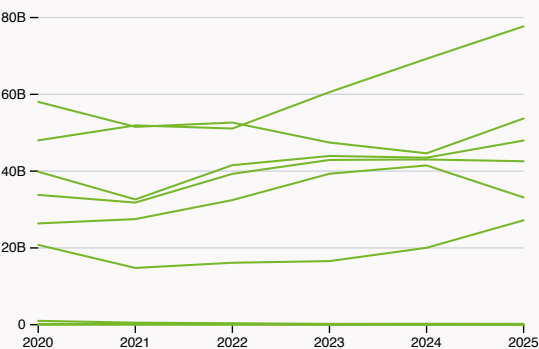
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Regional value



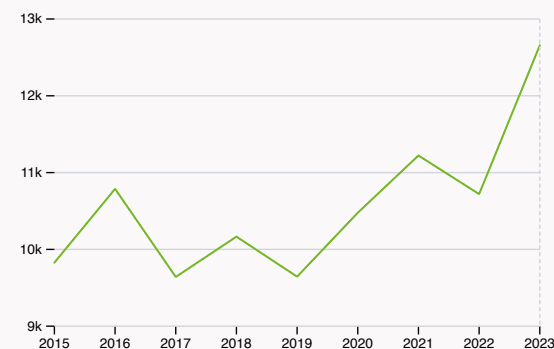
Economy values



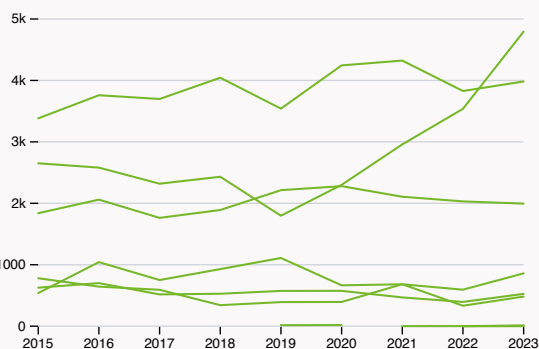
7.1.3 Global brand value, top 5,000

The value of the global brands from ASEAN in the top 5000 grew at an annual rate of 4% (CAGR) between 2020 and 2025, rising from USD 228.088 billion to USD 282.543 billion. In 2025, the value of the global brands from ASEAN in the top 5000 was up by 7.8 percentage points compared to the year prior.

Regional value



Economy values



7.1.4 Industrial designs by origin

The number of Industrial designs by origin in ASEAN grew at an annual rate of 3% (CAGR) between 2015 and 2023, rising from 10k industrial designs to 13k industrial designs. In 2023, Industrial designs by origin was up by 18.1 percentage points compared to the year prior.

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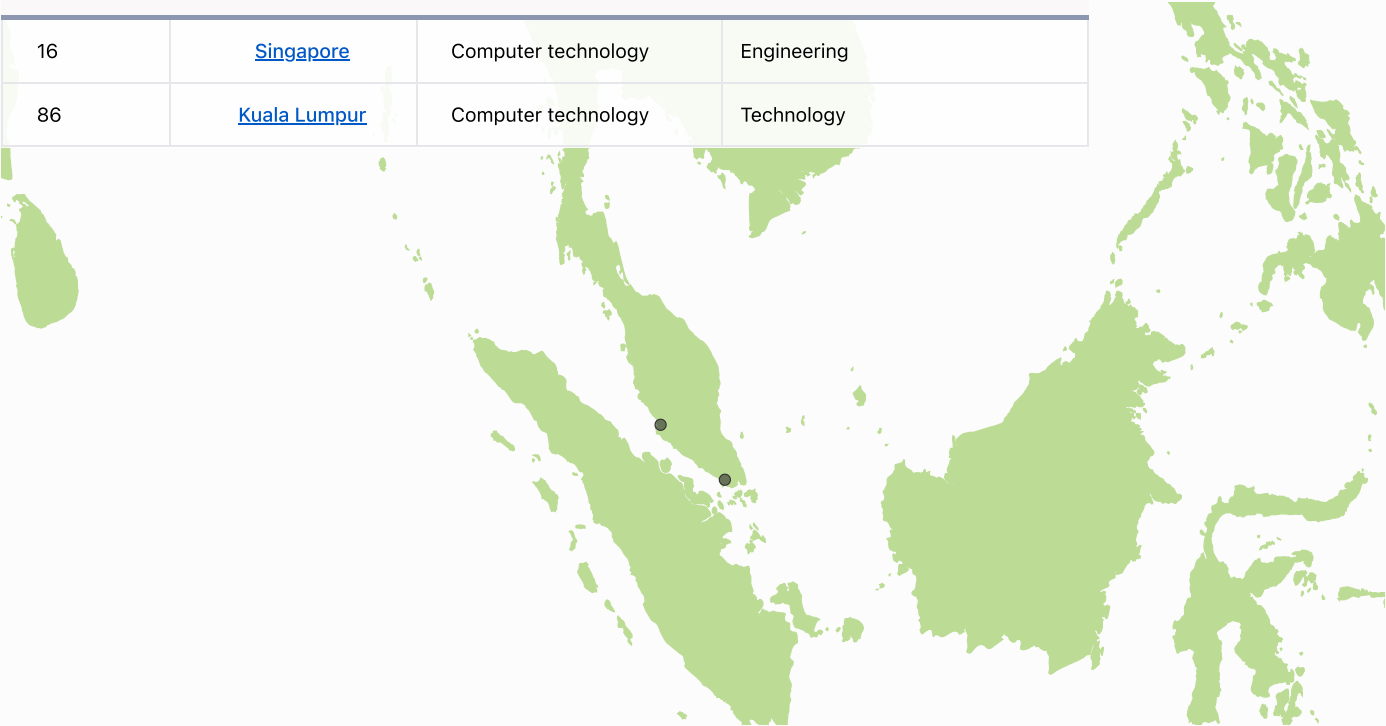
Top innovation clusters in ASEAN



ASEAN has 2 clusters in the world's top innovation clusters of the Global Innovation Index

The table and map below give an overview of the top innovation clusters in ASEAN.

Rank	Cluster name	Top patent field	Top academic subject
16	Singapore	Computer technology	Engineering
86	Kuala Lumpur	Computer technology	Technology

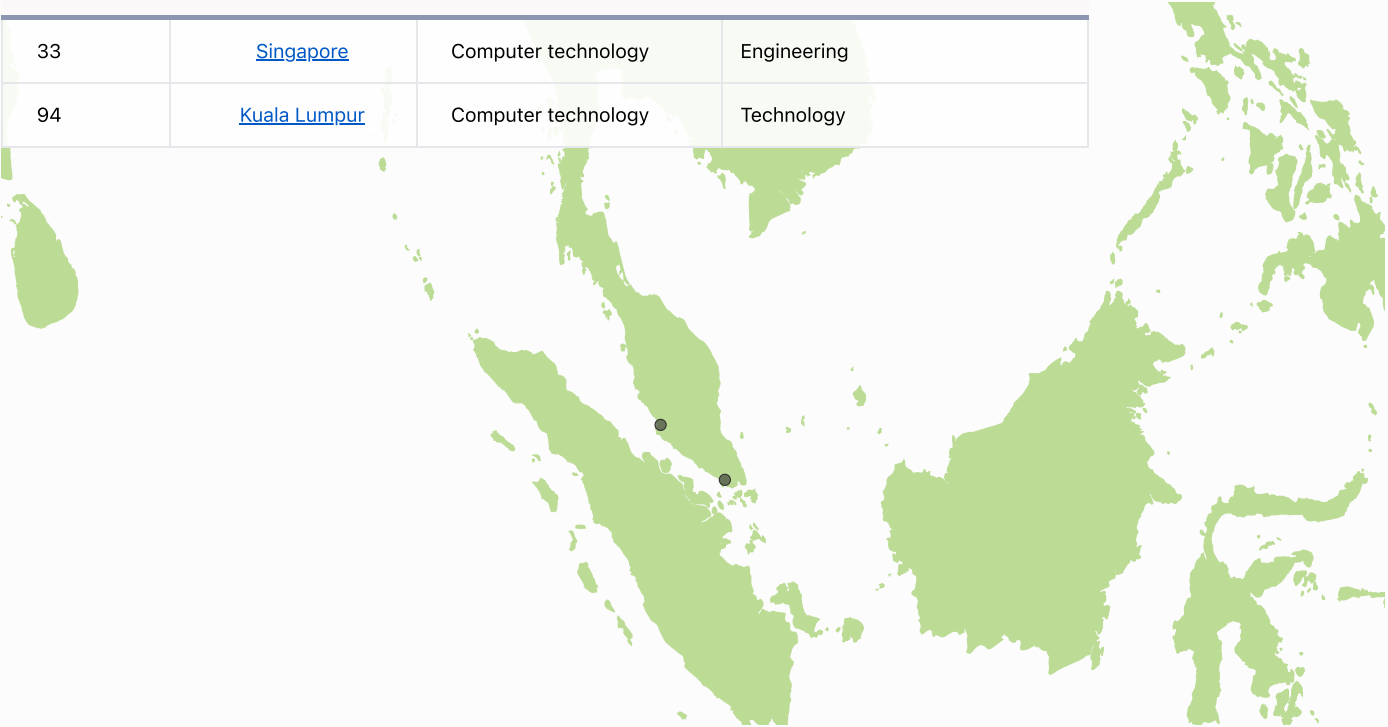


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The table and map below give an overview by intensity of the top innovation clusters in ASEAN.

Rank	Cluster name	Top patent field	Top academic subject
33	Singapore	Computer technology	Engineering
94	Kuala Lumpur	Computer technology	Technology



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ASEAN's Innovation Top Performers

Disclaimer: This section contains only the top performers by region or group. For the complete list, please visit the GII Innovation Ecosystems and Data Explorer website.

2.3.3 Global corporate R&D investors in ASEAN

World Rank	Region Rank	Firm	Country	Industry	R&D [mn EUR]	R&D Growth [%]	R&D Intensity [%]
217	1	SEA	Singapore	Software & Computer Services	1,049.02	n/a	8.82
326	2	UOL GROUP	Singapore	Real Estate Investment & Services	661.38	n/a	35.93
512	3	GRAB HOLDINGS	Singapore	Software & Computer Services	383.39	n/a	17.85

Source: The 2024 EU Industrial R&D Investment Scoreboard and Orbis database

Note: Data is based on the 2024 EU Industrial R&D Investment Scoreboard from the European Commission's Joint Research Centre, which ranks the top 2,000 firms by R&D investment annually. For countries not represented in the Scoreboard, companies from Orbis with R&D expenditure above USD 50 million were identified and used to complement the dataset.

2.3.4 QS university ranking of ASEAN's top universities

World Rank	University	Country	Score
8	NATIONAL UNIVERSITY OF SINGAPORE (NUS)	Singapore	93.7
15	NANYANG TECHNOLOGICAL UNIVERSITY, SINGAPORE (NTU)	Singapore	88.4
60	UNIVERSITI MALAYA (UM)	Malaysia	71.2

Source: QS Quacquarelli Symonds Ltd

Note: QS Quacquarelli Symonds Ltd annually assesses over 1,200 universities across the globe and scores them between [0,100]. Ranks can represent a single value 'x', a tie 'x=' or a range 'x-y'.

5.2.3 University industry and international engagement in ASEAN

Country Rank	University	Country	Score
1	NANYANG TECHNOLOGICAL UNIVERSITY, SINGAPORE	Singapore	97
2	NATIONAL UNIVERSITY OF SINGAPORE	Singapore	95.95
1	UNIVERSITI TEKNOLOGI PETRONAS	Malaysia	79.1

Source: Times Higher Education (THE), World University Rankings 2025.

Note: Rank corresponds to within economy ranks. The score is calculated as the average of the International Outlook score (encompassing international staff, students, and co-authorship) and the industry score (reflecting industry income and patent citations). The 2025 ranking corresponds to data from the academic year that ended in 2022.

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6.2.2 Top Unicorn Companies in ASEAN

World Rank ▼	Region Rank	Unicorn Company	Industry	Country	City	Valuation, bn USD
5	1	SHEIN	Consumer & Retail	Singapore	Singapore	66.00
235	2	HYALROUTE	Industrials	Singapore	Singapore	3.50
282	3	TRAVELOKA	Consumer & Retail	Indonesia	Jakarta	3.00

Source: CBInsights, Tracker , The Complete List of Unicorn Companies and Africa: The Big Deal

7.1.1 Top 15 intangible-asset intensive companies in ASEAN

Country Rank	Firm	Country	Intensity, % ^
9	METRO PACIFIC INVESTMENTS CORPORATION	Philippines	131.02
8	MALAYSIA AIRPORTS HOLDINGS BERHAD	Malaysia	118.94
6	SYNERGY GRID & DEVELOPMENT PHILS., INC.	Philippines	117.02

Source: Brand Finance Global Intangible Finance Tracker

Note: Brand Finance only provides within economy ranks.

7.1.3 Top 5,000 companies in ASEAN with highest global brand value

Country Rank	Brand	Country	Industry	Brand Value, mn USD
1	DBS	Singapore	Banking	17,217.22
1	PETRONAS	Malaysia	Oil & Gas	14,365.86
1	PTT	Thailand	Oil & Gas	9,169.80

Source: Brand Finance

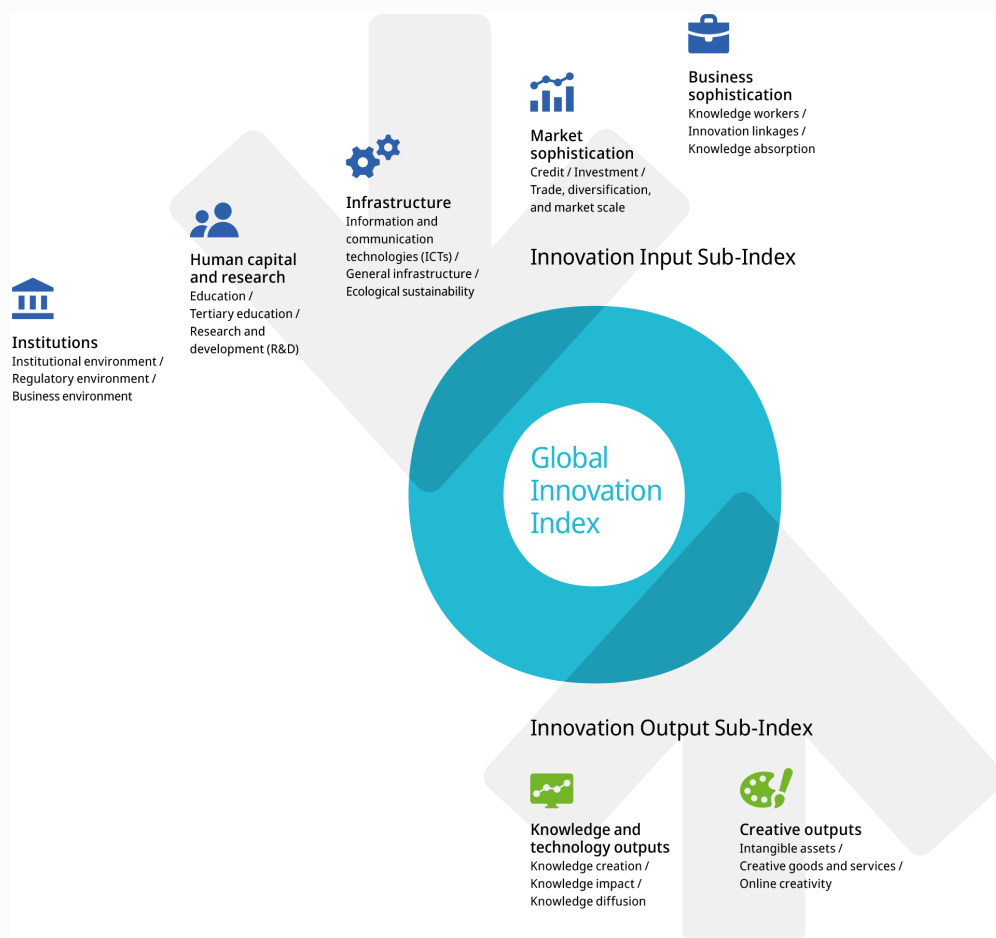
Note: Rank corresponds to within economy ranks.

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About the Global Innovation Index

- The Global Innovation Index (GII) is published by the World Intellectual Property Organization (WIPO), a specialized agency of the United Nations.
- Recognizing that innovation is a key driver of economic development, the GII aims to provide an innovation ranking and rich analysis referencing around 140 economies. Over the last decade, the GII has established itself as both a leading reference on innovation and a “tool for action” for economies that incorporate the GII into their innovation agendas.



The Index is a ranking of the innovation capabilities and results of world economies. It measures innovation based on criteria that include institutions, human capital and research infrastructure, credit, investment, linkages, the creation, absorption and diffusion of knowledge and creative outputs.

The GII has two sub-indices: the Innovation Input Sub-Index and the Innovation Output Sub-Index, and seven pillars, each consisting of three sub-pillars.